

The Half-Life of a Pricing Error

Trading the crypto weekend, 2017-2026: seven constructions, and why the fee schedule
beat all of them

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*Draft. Companion to **The Price of Calendar Time in a Market That Never Closes** (Paper I), which establishes the pricing error this paper tries to harvest. Every figure in the text is reproducible from scripts/weekend_*.py.*

Data provenance. All results are computed on verified series: BTC 24,349,954 trades, ETH 16,207,332, SOL 695,295, XRP 234,640, with bar coverage at 100%, 100%, 99.9% and 99.9%. Three bugs were found and fixed during this draft, one of them in this paper's own trading engine — a look-ahead condition that made the exit mark conditional on the option still being near the money, deleting the weekends on which the index moved and roughly doubling the headline result. All three are documented in docs/data_notes.md, with the superseded claims recorded in the sections they affected.

A fourth bug, found after that list was written, is the reason this draft exists in its present form. The vega-matched spread was constructed as `net_weekend - net_weekday`. Both stored legs are the P&L of a *short* contract and the engine always subtracts costs, so negating the second leg turned its costs into a credit. An implementable position that is short one contract and long another pays the costs of both. The reported edge was overstated by exactly twice the long leg's cost — 0.064 per unit vega at daily rehedgeing and 0.267 at five-minute, against a reported edge of 0.039 and 0.021. **The spread was never profitable net of costs. It has not decayed; it was**

never there. Every P&L number below is the corrected one, the superseded figures are stated where they stood, and §12 is new.

Abstract

Paper I documents a persistent pricing error in Deribit's crypto options: the market prices the weekend as though it were an ordinary stretch of trading time, when Saturday realizes about a third of a weekday's variance, and the implied discount has been deepening for a decade. This paper asks the natural follow-up. **Can it be harvested, and if it once could, what happened?**

The answer is a null, and a sharper one than an earlier draft of this paper claimed. The vega-matched calendar spread that isolates the error has a *gross* edge of +0.042 per unit vega in both mature books, which tracks the pricing gap across all four underlyings with the right sign and the right rank. It has never covered its costs. Charging both legs — as an implementable position must, and as an earlier version of this paper did not — leaves **-0.025 (t = -2.09) in Bitcoin and -0.029 (t = -1.82) in Ether over the full history**, at the cheapest rehedging frequency available. Stripping the trade back to a single outright short, where no such construction is possible, earns +0.043 per unit vega over the full history and **-0.001 (t = -0.12) over the last two years**.

Along the way the paper measures the anatomy of the trade rather than merely its sign. Selling weekend content pays (+0.127 per unit of weekend share in the life sold) and holding through the weekend costs (-0.080 per unit of the window), because implied volatility mechanically re-rates upward as weekend-heavy remaining life becomes weekday-heavy and a short pays that on the mark. A second-order attribution locates the entire cost in the vega term and finds the quiet weekend arriving, as it should, as a *credit* in the gamma term. Of eight pre-specified conditioning variables only one survives — the ratio of the entry quote to the volatility the week has already realized — and it raises the out-of-sample hit rate by sixteen and fourteen points without rescuing a losing period. Across the moneyiness ladder there is nowhere better to run the trade than at the money, and the in-the-money side cannot be traded at size at all.

The commercial conclusion is unambiguous and negative, and the correction sharpens rather than softens it. Because the binding constraint is cost rather than signal, the paper prices the one configuration in which the trade could work: a market maker, who earns the effective half-spread instead of paying it on both legs. That is worth four half-spreads, and it closes two thirds of the gap without crossing zero — **-0.008 in Bitcoin and -0.004 in Ether**. Splitting the residual shows why: of 0.058 per unit vega in fees, **exchange option fees are 0.030 and perpetual hedging is 0.028**. A maker on a fee tier that discounts option fees by 27% in Bitcoin or 12% in Ether clears zero on full-history data at roughly +0.022. **Over the last twelve months even that configuration loses 0.013**. The pricing error in Paper I is real and is not going away; no construction available on this venue has ever monetised it net of costs, and the fee schedule is the reason.

1. Introduction

The natural objection to a systematic pricing error is that it should be arbitrated away. This paper is an attempt to do exactly that, reported in full including the parts that failed, and its interest is in the shape of the failure rather than in the failure itself.

Three things make the exercise worth reporting rather than filing. First, the decay is *measurable*: the same trade, on the same construction, run date by date across nine years, goes from a Sharpe above one to an inversion, and the date at which it turns is informative about when the market started pricing the weekend properly. Second, a trading P&L is the strongest available validation of a measurement chain — a pricing gap that cannot be shorted at all is a gap that might be an artefact of the estimator, and Paper I's gap could be shorted, profitably, for years. Third, the anatomy of the

trade turns out to identify the mechanism more sharply than the quotes do: the P&L separates selling weekend content from living through the weekend, which the quote-side regressions in Paper I cannot.

The paper also reports two of its own errors, because the corrections change conclusions rather than decimal places. One is a look-ahead condition in the trading engine described in the provenance note above. The other is a discarded finding that returned once that condition was removed. Both are flagged where they occur.

2. The trade, and how its costs are measured

The natural objection to a systematic pricing error is that it should be arbitrated away. Here it partly has been.

The trade that isolates the effect is a vega-matched calendar spread: sell the weekend-heavy contract, buy the weekday-heavy one, delta-hedge both in the perpetual, hold to settlement. Buckets are assigned within each day's own cross-section — an absolute threshold makes the legs nearly mutually exclusive, since Fridays offer weekend-heavy contracts and Tuesdays offer weekday-only ones, so a fixed cutoff describes a spread that could rarely be put on.

Costs are measured rather than assumed. Deribit charges 0.03% of the underlying on options, capped at 12.5% of premium, plus 0.05% taker on the perpetual hedge at every rebalance. The effective half-spread, recovered from the tape by differencing buyer-paid against seller-received implied volatility on the same instrument-day, is **0.42 volatility points**.

Bitcoin, 1,231 paired days, per unit of vega:

Rehedge interval	Level (gross)	Level (net)	Spread (net)	t	Sharpe
5 minutes	+0.0445	−0.0896	+0.0209	3.83	1.73
1 hour	+0.0539	−0.0038	+0.0160	2.51	1.13
8 hours	+0.0614	+0.0244	+0.0288	3.19	1.44
daily	+0.0583	+0.0251	+0.0391	3.33	1.50

Two readings, and the second is the finding.

The level column validates the engine. A short delta-hedged option book earns a positive premium gross at every frequency, as the variance-risk-premium literature requires. It turns negative net only under five-minute rebalancing, where perpetual taker fees are paid some two thousand times over a seven-day contract and cost 0.134 per unit vega — more than the premium is worth. That is a statement about an unrealistic hedging rule, not about the market.

The spread trade is profitable at every frequency, and most profitable where hedging is cheapest: at daily rebalancing it earns 0.039 per unit vega net of measured fees and spread, $t = 3.33$, annualized Sharpe **1.50**.

3. The P&L tracks the pricing gap in all four assets

The trade sells weekend-heavy variance, so it should profit where weekends are priced rich and lose where they are priced cheap. Comparing against Paper I §5.1's gaps, at five-minute rehedge — the setting the cross-asset stage runs at, and the most conservative of the four in the table above because it pays the most perpetual fees:

Asset	Pricing gap	Spread P&L, gross	Paired days
BTC	+0.051	+0.0226	1,231
ETH	+0.038	+0.0212	1,213
SOL	-0.099	-0.0004	377
XRP	-0.133	-0.0254	284

This test is stated on gross P&L, and deliberately. Costs are a property of the venue's fee schedule, not of the weekend, so a validation of the *measurement* should not be routed through them; §4 charges them in full and the answer there is negative in every book. Read gross, every sign matches and the estimates are **rank-ordered perfectly**: sorting the four books by their implied-versus- realized pricing gap sorts them by gross trading profit, from Bitcoin's +0.051 and +0.0226 down to XRP's -0.133 and -0.0254. The ordering is carried by point estimates rather than four independent rejections — but the ordering itself is the prediction, and it holds without exception.

An earlier version of this table reported *net* P&L of +0.0209, +0.0151, -0.0103 and -0.0291, and read the rank ordering off those. Those figures credited the long leg's costs rather than charging them; the ranking happened to survive the correction because the bias is a similar size in every book, but the levels did not, and none of the four is positive net of costs at this rehedging frequency.

This is the paper's strongest internal check. The two legs differ in maturity as well as weekend coverage, so a mechanical bias — short-dated contracts systematically out-earning longer-dated ones — would generate same-signed profits in every asset regardless of its pricing gap. Instead the sign, and now the rank, tracks the gap across four independent books with two different settlement conventions. **Implied-volatility gaps measured from quotes translate into realized trading profits with the correct sign and the correct order**, which validates the measurement chain far better than a single profitable backtest.

Note also that Ether's gross level is now positive (+0.0295 per unit vega), as the variance risk premium requires. An earlier version of this section reported it at -0.176 and treated the anomaly as an open puzzle; it was the corrupted bar series, whose gaps sent the hedging path through price jumps that never occurred. That diagnosis is what led to the bar-coverage guard now in `bars.load`.

3.1 The edge is being competed away, in every book

Splitting each sample in half:

Asset	First half	t	Second half	t
BTC	+0.0343	3.80	+0.0076	1.24
ETH	+0.0465	3.19	-0.0162	-2.20
SOL	+0.0018	0.09	-0.0224	-1.69
XRP	-0.0128	-0.42	-0.0454	-2.87

Bitcoin's edge has roughly quartered and lost significance. Ether's has *reversed* — significantly negative in the second half. Every one of the four books has a second half worse than its first, and the two that are significant in the second half are both significantly negative.

That decay is the more interesting result, and it replicates in all four. It is a statement about the *gross* signal narrowing, not about a profit being competed away — §4 shows there was never a net profit to compete away — and on that reading it still qualifies every full-sample pricing estimate in Paper I §5: they average over a period during which the weekend error was shrinking, so the *current* mispricing is smaller than the pooled figures suggest. What the narrowing cannot be is evidence of arbitrage capital entering this trade, since the trade did not pay. Someone is quoting against the error; they are not doing it by crossing the spread.

The two young books deserve a caveat rather than the same reading. Solana and XRP were listed in 2024, so their “first half” begins where Bitcoin’s second half is already well advanced; their decline is a decline within a late window, not a replication of the same multi-year arc. What the four rows share is a direction, not a common clock.

A note on two earlier versions of this section. Before the index-alignment bug was found, this analysis reported the trade as unprofitable and *undetectable* — a per-trade dispersion of 5.0 per unit vega against an implied edge of 0.031 — which invited a limits-to-arbitrage reading. Corrected, the dispersion is 0.19. Then, before the bar-coverage bug was found, it reported Ether’s spread at -0.164 and presented that as confirming a sign prediction derived from Ether’s corrupted gap of -0.205 . Both the prediction and the test used corrupted inputs, and the apparent agreement — $4.2\times$ observed against $4.0\times$ predicted — was two errors cancelling. The table above is the version computed on verified data.

4. The spread never covered its costs

The half-sample split above ends on a direction rather than a level, and a direction is not enough to answer the question a desk would ask: is the trade on today? Charged correctly, the answer is that it was never on. Walking the rehedging interval out to the frequency where costs are lowest, and reporting gross alongside net so the two can be told apart:

BTC	5 min	60 min	8 hours	daily
gross	+0.0226	+0.0192	+0.0311	+0.0418
fees, both legs	0.2599	0.1071	0.0658	0.0579
spread	0.0083	0.0083	0.0083	0.0083
crossing, both legs				
taker net	-0.2457	-0.0962	-0.0430	-0.0245
<i>as previously reported</i>	<i>+0.0209</i>	<i>+0.0160</i>	<i>+0.0288</i>	<i>+0.0391</i>

Ether is the same shape: gross $+0.0418$ at daily rehedging against 0.0581 of fees and 0.0127 of spread, for a net of **-0.0289** . Solana and XRP have negative gross to begin with and are worse at every rung.

The signal is real and smaller than the toll. Bitcoin’s gross edge of 0.042 per unit vega is a genuine measurement of weekend richness — it is what §3’s rank ordering rests on — but the venue charges 0.066 to collect it. Nothing about the weekend changed; the arithmetic was wrong. Every superseded figure is shown in italics above.

The rest of this section is retained because the *time profile* is unaffected: the bias is a near-constant per-day charge, so it shifts the level of every subsequent table without changing its slope. Refitting date by date, with five-minute rehedging and the previous costing, and reading it by calendar year:

	BTC	ETH	SOL	XRP
2020	+0.0192 (+0.9)	+0.0993 (+2.0)		
2021	+0.0596 (+2.8)	+0.0617 (+2.3)		
2022	+0.0414 (+3.0)	+0.0404 (+2.3)		
2023	+0.0172 (+1.8)	-0.0064 (-0.6)		
2024	+0.0212 (+1.8)	-0.0177 (-1.6)	+0.0151 (+0.6)	+0.0026 (+0.1)
2025	+0.0117 (+1.1)	-0.0101 (-0.6)	-0.0294 (-1.7)	-0.0367 (-1.6)
2026	-0.0367 (-3.1)	-0.0456 (-3.4)	-0.0094 (-0.5)	-0.0485 (-2.2)
last 12m	-0.0188 (-2.2)	-0.0301 (-2.7)	-0.0127 (-0.8)	-0.0393 (-2.4)
last 6m	-0.0428 (-3.3)	-0.0549 (-3.7)	-0.0311 (-1.5)	-0.0636 (-2.9)
full	+0.0209 (+3.8)	+0.0151 (+1.8)	-0.0103 (-0.8)	-0.0291 (-1.7)

Mean P&L per unit vega with t -statistics beside it. **The trade has gone from decayed to inverted.** Over the last twelve months it loses money in all four books and does so significantly in three; over the last six it loses in all four at -0.04 to -0.06 per unit vega, against a full-sample Bitcoin edge of $+0.021$. The hit rate falls with it, from 63% in Bitcoin's best year to 40% over the last six months. Selling the weekend against the weekday is now a losing trade at every horizon short enough to describe current conditions.

This is not a tail accident. Bitcoin's median day over the last twelve months is -0.008 and Ether's is -0.010 , so the centre of the P&L distribution has crossed zero on its own; removing the five worst days from each still leaves -0.011 and -0.019 . The tails make the loss worse — the distribution is left-skewed, at -0.29 in Bitcoin and -0.63 in Ether, and Bitcoin's five worst days all fall in February and March 2026 — but they are not what put it below zero.

Figure 11 shows the cumulative path. Ether's equity curve peaks in late 2022 and has given back a third of its gains; Bitcoin's runs to a high in early 2025 and turns over. Solana and XRP never establish a positive stretch at all.

All of this is measured at the five-minute rehedge used elsewhere here. §5 walks that interval out and finds the inversion does not survive to daily rebalancing — not because the sign flips, but because discrete-hedging error more than doubles the series' standard deviation and swamps an effect this size. The claim is therefore that the trade has inverted *at a rehedge frequency fine enough to see it*, which is the frequency this paper's bar data supports and the one at which the full-sample edge was established in the first place.

The reason is the same one Paper I §5.6 identified, seen from the seller's side. Comparing the implied volatility at which Friday's weekend-covering contracts actually traded against what the weekend then delivered:

	2020	2021	2022	2023	2024	2025	2026
BTC median	+0.081	+0.169	+0.154	+0.094	+0.125	+0.065	+0.035
BTC mean	+0.037	+0.080	+0.129	+0.086	+0.104	+0.052	+0.006
ETH median	+0.149	+0.161	+0.138	+0.071	+0.086	+0.094	+0.067
ETH mean	+0.046	+0.080	+0.090	+0.053	+0.069	+0.041	+0.015

In annualized volatility units. The seller's cushion has compressed by an order of magnitude in Bitcoin since 2022, and the mean is now within a rounding error of zero while the median is still visibly positive. That gap between the two columns is the whole story: a desk reading a screen sees a weekend still quoted above what weekends usually deliver and concludes the sale is on, while the quantity it is actually paid on — the mean — has already gone. Add two crossings of a 0.42-vol-point spread and the transaction costs of a hedged position, and what is left is negative.

So the market's deepening discount and the disappearance of the trade are two readings of one fact. The discount had somewhere to go because the weekend's typical variance really was falling; the trade stopped working because the weekend's *mean* variance stopped falling with it. Whoever is still shorting weekend variance on a screen-read edge has been paying for the difference since roughly the start of 2025.

Figure 11. The trade has inverted, and the seller's cushion is gone. Left: cumulative P&L per unit vega of the vega-matched spread that sells the weekend-heavy contract and buys the weekday-only one, net of measured exchange fees, perpetual rehedge costs and two crossings of the 0.42-vol-point effective spread. Bitcoin and Ether in bold; Solana and XRP, listed in 2024, faded.

Right: the implied volatility at which Friday's weekend-covering contracts traded, minus the volatility the weekend then realized, by year. Median (solid) against mean (dashed): the gap between them is the moment mismatch of Paper I §5.6, and the mean reaching zero is what ends the trade.

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5. Dropping the weekday leg

The weekday leg exists to difference out everything that is not the weekend, and it is fair to ask what it costs. Selling weekend-heavy contracts outright and skipping the hedge is simpler, ties up less margin, and pays one spread instead of two. The answer is that it changes the trade into a different trade, and the change has to be measured at a rehedg frequency where the comparison is even possible.

At five-minute rehedge the outright short cannot be evaluated at all. It loses 0.079 per unit vega in Bitcoin and 0.102 in Ether, but 0.135 and 0.143 of that is perpetual taker fees. The two legs of the spread carry similar vega and therefore similar rebalancing costs, so the fee drag largely differences out; standing alone, the weekend leg pays all of it. That is a fact about the rebalancing rule, not about weekends. Walking the interval out:

	5 min	60 min	8 hours	daily
BTC fee drag	0.135	0.059	0.038	0.034
BTC outright, net	-0.079 (-16.5)	+0.004 (+0.7)	+0.039 (+4.8)	+0.045 (+4.5)
BTC spread, net	+0.021 (+3.8)	+0.016 (+2.5)	+0.029 (+3.2)	+0.039 (+3.3)
ETH outright, net	-0.102 (-16.0)	-0.017 (-2.3)	+0.011 (+1.0)	+0.010 (+0.7)
ETH spread, net	+0.015 (+1.8)	+0.012 (+1.4)	+0.013 (+1.1)	+0.038 (+2.4)

Full sample, per unit vega, *t*-statistics beside. Daily rebalancing is the only rung at which an outright short is a real proposition, so it is the one to read.

There the outright looks better in Bitcoin and worse in Ether, which is the tell. Bitcoin's outright earns +0.045 at an annualized Sharpe of 2.03 against the spread's +0.039 at 1.50; Ether's earns +0.010 at 0.33 against the spread's +0.038 at 1.10. A weekend effect would not reverse between two books that both price the weekend the same way. A variance risk premium would, and does: Bitcoin's outright gross level is +0.079 and Ether's is +0.047, and Solana's is +0.000 — Solana has essentially no variance premium in this sample, and its outright short is negative at every rung, in every year.

So the extra return in Bitcoin is not weekend alpha. It is the variance risk premium, which the weekday leg was never there to capture and which a *weekday* short would have earned just as well. **Re-**

moving the hedge does not sharpen the weekend bet; it buries it inside a much larger short-volatility position. The correlation between the two series is 0.57 in Bitcoin and 0.61 in Ether, so most of what the outright does day to day is something else entirely.

The risk is worse in the way that matters for a short-volatility book. Ether's outright draws down 21.3 per unit vega at its worst against the spread's 9.5, and its worst sixty-day stretch is -8.5 against -7.4. Bitcoin's are comparable in this sample — 6.9 against 6.7 — but Bitcoin is the book whose realized weekend tail has grown least, and Paper I §5.6 shows that tail growing in both.

And it does not rescue the trade. By year, at daily rehedging:

	BTC	ETH	SOL	XRP
2021	+0.125 (+3.2)	+0.091 (+1.8)		
2022	+0.094 (+3.3)	+0.040 (+1.0)		
2023	+0.039 (+1.8)	+0.007 (+0.3)		
2024	+0.023 (+1.2)	+0.001 (+0.0)	+0.007 (+0.1)	-0.111 (-1.1)
2025	-0.011 (-0.6)	-0.051 (-1.4)	-0.080 (-1.6)	-0.019 (-0.3)
2026	-0.013 (-0.7)	-0.032 (-1.2)	-0.020 (-0.4)	+0.039 (+1.1)
last 12m	-0.017 (-1.1)	-0.023 (-0.9)	-0.033 (-0.8)	+0.042 (+1.3)

The same arc as §4: strong through 2022, decayed by 2024, negative in three of four books over the last twelve months. The outright version is noisier — none of the recent estimates is individually significant — so it fails to reject rather than confirming, but it certainly does not point the other way. Dropping the weekday leg trades a small measurable edge that has gone for a large unmeasurable one that has also gone.

A caveat this ladder forces on §4. The recent inversion of the spread is measured at five-minute rehedging, and it does not survive to the coarse end of the ladder: Bitcoin's last-twelve-month spread runs -0.019 ($t = -2.2$) at five minutes, -0.019 ($t = -2.0$) at an hour, -0.010 at eight hours and +0.004 at daily. That is a power loss rather than a sign reversal — the standard deviation of the same series rises from 0.125 to 0.312 as discrete-hedging error takes over, so the daily estimate's standard error is 0.022 and every point estimate in the row sits inside one of them. It is the same reason §6 hedges at five minutes in the first place. But it should be stated plainly: **the inversion is established at fine rehedging and is not independently significant at coarse rehedging**, and a desk that rebalances daily would not have been able to detect it.

6. Trading the clock instead of the contract

Everything above enters when a trade happens to print and holds to settlement. Settlement is the right exit for measuring a pricing error, because it is the one valuation that is not a matter of opinion, but it is not how a desk would run the position. The clock version sells the most weekend-heavy contract available at a fixed hour on Friday and buys it back at a fixed hour later, marked at whatever the market was quoting then. Two things get worse — the spread is crossed twice instead of once, and the exit mark is an opinion — and one thing gets better: the hold is thirty-five hours rather than a week, so far less is paid to hedge it.

A correction to an earlier version of this section. The first implementation looked for the exit mark in the same frame it drew entries from, and that frame had already been narrowed to the 0.35–0.65 delta band. The exit mark was therefore conditional on the option still being near the money when the position was closed — a condition on the future. Options leave the band when the index moves, and a delta-hedged short loses when the index moves, so the requirement deleted losing weekends: the Fridays that survived it had a mean absolute weekend index move of 0.94% against 2.15% across all Fridays, and within the surviving Fridays it forced a different contract on 63 of 82 occasions, always the one that had stayed still. It doubled the reported result. The exit index is now

built from an unbanded frame — the delta band constrains what may be *entered*, nothing more — and the numbers below are the corrected ones. The figures in the previous draft (+0.094 and +0.095 at hourly rehedge) should be read as roughly twice their true size. The bug was visible in the earlier draft’s own limitations paragraph, which noted that only 82 of some 430 Fridays filled and that “which fifth fills is not random”; the fill is now 227 of 496 in Bitcoin and 158 of 386 in Ether.

Entering at 12:00 UTC on Friday and closing at 00:00 UTC on Sunday, which is the first instant after Saturday has finished:

rehedge	BTC net	t	hit	ETH net	t	hit	cost
5 min	-0.036	-3.40	43%	-0.057	-3.79	37%	0.102 / 0.126
30 min	+0.020	1.76	58%	+0.018	1.17	61%	0.060 / 0.073
60 min	+0.043	3.91	64%	+0.038	2.40	63%	0.051 / 0.063
4 hours	+0.069	5.26	69%	+0.074	4.22	70%	0.041 / 0.051

Per unit vega, 227 Bitcoin and 158 Ether trades. **At the five-minute rehedge \$6 uses, the trade loses in both books:** perpetual fees alone take more than the position earns. Five minutes is the right frequency for a seven-day hold whose discrete-hedging error would otherwise swamp the signal; it is the wrong frequency for a thirty-five hour one, where there is far less gamma risk to hedge away and the same fee schedule applies. At hourly rebalancing both books clear, by four and two and a half standard errors.

Unlike the previous draft, gross P&L is no longer flat across the rungs — it runs +0.066 to +0.110 in Bitcoin as rehedge coarsens. That is the discrete-hedging error of \$5 reappearing, and it is visible now precisely because the moving weekends are back in the sample: on a quiet path the rehedge frequency barely matters, and the earlier version had kept only quiet paths.

This remains the most profitable version of the weekend trade in the paper, and the reason is not a better view. It is a shorter hold over the specific stretch that Paper I §3 shows is quietest, paying for a hedge over thirty-five hours instead of seven days. But it is now a trade whose sign depends on the hedging frequency, which is a much weaker claim than the one this section previously made.

Is the frequency dependence about the weekend, or about not hedging? A short option hedged coarsely profits when the path mean-reverts, because it avoids buying high and selling low between rebalances. Quiet weekends mean-revert, so the ladder above could be harvesting weekend mean reversion rather than the weekend variance mispricing — a different trade with a different risk. The test is to run the same ladder on an exit that holds *no* weekend, on a common contract menu:

rehedge	BTC gross, no weekend	BTC gross, holds Saturday	ETH gross, no weekend	ETH gross, holds Saturday
5 min	+0.015	+0.066	+0.018	+0.069
30 min	+0.019	+0.080	+0.026	+0.091
60 min	+0.026	+0.094	+0.027	+0.101
4 hours	+0.030	+0.110	+0.032	+0.125
gain from coarsening	+0.015	+0.044	+0.013	+0.056

Both answers are partly true, and the split is informative. Coarsening helps the weekend-free exit too, by about 0.014 in both books — that is the generic mean-reversion effect, and it is real. But it helps the weekend-holding exit three to four times as much, and **at every rehedge frequency,**

including the finest, the exit that holds a Saturday earns four to five times the gross of the one that does not. The weekend effect exists independently of how often the hedge is rebalanced; coarsening amplifies it rather than creating it, which is what one would expect if the weekend path is the calmer one. The no-weekend exit never turns positive net of costs at any frequency, reaching only -0.009 . What makes this trade profitable at all is the Saturday.

Which midnight matters. Closing at 00:00 UTC on *Saturday* — the start of Saturday rather than the end — holds no weekend at all, only Friday afternoon. At five-minute rehedge it runs -0.049 ($t = -5.60$) in Bitcoin and -0.060 ($t = -5.25$) in Ether, against -0.036 and -0.057 for the Sunday-midnight exit. Both lose at that frequency, so this comparison is now about which loses less, and the ordering is weak.

A pattern that was discarded, and now returns. The previous draft found that extending the exit to Monday 00:00 looked much worse, called it an artefact of contract selection, and reported it as discarded. Under the corrected exit index it survives the control that killed it. Restricting every exit to contracts that live to the latest of them, so all four trade from the same menu on the same Fridays, Bitcoin runs -0.024 , -0.006 , -0.028 and **-0.068** ($t = -5.37$) across the Saturday-start, Saturday-noon, Sunday-start and Monday-start exits on 86 common Fridays, and Ether -0.037 , -0.021 , -0.034 and **-0.092** ($t = -5.28$) on 50. The Monday exit is the worst by a wide margin on a menu that holds the instrument fixed. What separates it is not the realized path but the mark: implied volatility falls 1.3 volatility points over the hold to Sunday midnight and *rises* 6.2 points over the hold to Monday. That is the re-rating mechanism of §8 — as weekend-heavy remaining life becomes weekday-heavy, the quote goes up, and the short pays for it — showing up in the P&L of a trade that was never designed to test it. **Holding a weekend short into Monday gives back more than the weekend earned.**

Two limits. Requiring a print within forty-five minutes of Sunday midnight still fills fewer than half the Fridays, and which half fills is not random — a contract trading at that hour may be one where something is happening. Widening the window to four and eight hours raises the Bitcoin sample to 244 and 266 and moves the five-minute result from -0.036 to $+0.004$ and $+0.017$, so the window matters and the direction of the selection is now measurable rather than assumed. And this is an outright short with no weekday leg, so by §5 it collects some variance risk premium as well as any weekend effect; its level is not a clean estimate of the weekend error. Solana and XRP fill seven and four Fridays and are reported only to record that their books cannot support a fixed-hour rule at all.

7. Can the losing weekends be identified in advance?

The clock trade of §6 loses on roughly a third of weekends. A desk would want to know which third. With 244 Bitcoin trades and a free choice of conditioning variables, however, something will always appear to work, so the exercise is only worth running under discipline: eight candidate factors fixed in advance with the sign each is expected to take, every one reported, Ether treated as a replication rather than a second attempt, a Bonferroni threshold for eight tests ($|t| \geq 2.73$), and a filter built on the first half of the sample and evaluated on the second.

factor	expected	BTC β /sd	t	ETH β /sd	t
IV over realized vol of the week so far	+	+0.032	3.71	+0.042	3.05
IV percentile in its own trailing year	+	+0.024	2.24	+0.026	1.80

factor	expected	BTC β /sd	t	ETH β /sd	t
entry implied volatility, level	+	+0.017	1.28	+0.026	1.37
weekend share of the contract's life	+	+0.013	1.34	+0.025	0.83
5-day change in DVOL	–	+0.016	0.60	+0.028	0.50
absolute index move over Friday	–	–0.002	–0.20	–0.020	–1.20
realized vol of the week so far	–	–0.005	–0.35	+0.008	0.36
absolute perpetual funding, trailing week	–	–0.001	–0.05	–0.007	–0.38

Slopes are P&L per unit vega per standard deviation of the condition, with heteroskedasticity-robust t . **Exactly one factor survives its sign and the Bonferroni threshold, and it does so in both books: the seller's cushion of Paper I §5.6** — the entry quote divided by the volatility the week has actually realized so far. It is +0.032 ($t = 3.71$) in Bitcoin and +0.042 ($t = 3.05$) in Ether, and it holds its sign and significance on the first half of each sample alone ($t = 3.32$ and 2.14). Everything else fails, though only the DVOL change fails with the wrong sign in both books, and it is estimated on the shortest sample here.

This reverses the previous draft, and the reason is instructive. Under the biased exit index of §6 the cushion was the *weakest* thing tested ($t = -1.26$ in Bitcoin) and perpetual funding was wrong-signed in both books. That is what a sample with the losing weekends deleted from it looks like: a factor whose job is to predict losses has nothing left to predict, and the residual variation it fits is noise. Restoring the moving weekends restored the variation, and the most theoretically motivated candidate on the list is the one that came back.

One factor remains a trap, even though it no longer looks impressive. The level of implied volatility is insignificant in both books here ($t = 1.28$ and 1.37), but it would still be the wrong thing to build a threshold on: implied volatility falls across the sample, which is Paper I §5.5's finding, so a rule denominated in volatility points is fitted to a level that does not persist. The percentile version in the table is the same idea made scale-free, and the out-of-sample filter below uses only scale-free factors for that reason.

One factor is maturity in disguise. With entry fixed at Friday noon, the weekend share of a contract's remaining life is a deterministic decreasing function of its maturity — the correlation is -1.00 — and the available maturities take two values, about 1.7 days and about 6.8. The factor is therefore not measuring weekend content at all at this entry point; it is saying that the short-dated contract pays more. §8 takes this apart by varying the entry day, and finds that the weekend share was not simply standing in for maturity: maturity carries its own effect, but the weekend measures keep their size beside it, and what a Friday-noon entry cannot separate is two weekend effects of opposite sign.

The filter, built on the first half and applied unseen to the second. Scale-free factors only, which selects the cushion plus the weekend share in Bitcoin and the cushion alone in Ether:

	trades	mean	hit rate	<i>t</i>
BTC, all	122	+0.002	51.6%	
BTC, filtered	34	+0.037	67.7%	1.82
ETH, all	84	-0.035	44.1%	
ETH, filtered	24	-0.028	58.3%	-0.74

On the question as posed the answer is now a qualified yes, where the previous draft's answer was no. The filter raises the hit rate by sixteen points in Bitcoin and fourteen in Ether, out of sample, on weekends it has never seen — the previous draft found no hit-rate improvement at all, again because the sample it was fitted to had few losses in it to screen out. It is genuinely identifying weekends the trade should be skipped on.

Two things keep this qualified rather than clean. Bitcoin's filtered mean is +0.037 on 34 trades with *t* = 1.82, which is suggestive and not significant at conventional levels. And Ether's out-of-sample half loses money outright, at -0.035 per unit vega across all trades; the filter improves it to -0.028 and lifts the hit rate, but improving a losing period is not the same as making it profitable. The honest summary is that the cushion carries real information about which weekends to skip, that it survives a pre-specified test and a split-half in both books, and that it is not large enough to rescue a period in which the underlying trade does not work.

8. Weekend content against maturity, separated

§7 could not tell the two apart. With entry pinned to Friday noon the weekend share of a contract's remaining life is a deterministic decreasing function of its maturity — the correlation was exactly -1.00 — and only two maturities were ever available, so “sell more weekend” and “sell less time” were the same instruction.

Breaking that needs the identification Paper I §4 already uses for quotes, applied to P&L. Deribit lists daily expiries, so weekend share is *not* monotone in maturity within a single instant: from a Wednesday, the contract expiring Thursday carries no weekend, the one expiring Monday carries a large share, and the one expiring the following Friday carries a smaller share again on a longer life. Entering on **every day of the week** rather than only Friday adds the second source of variation, because the same maturity then carries different weekend content depending on the day it is bought.

The design: enter at 12:00 UTC on every calendar day, sell the at-the-money contract at each available expiry between 0.6 and 14 days, hold a fixed thirty-six hours, re hedge hourly, mark out against a traded print. Holding the horizon fixed matters — it means the entry day alone moves the weekend content of the holding window. That gives 4,635 Bitcoin and 3,771 Ether trades, and it takes the raw correlation between weekend share and maturity from -1.00 to **-0.08** and **-0.10**.

Two quantities now separate, and they turn out to have **opposite signs**:

	weekend share of <i>life sold</i>	weekend share of <i>window held</i>	maturity, days
BTC	+0.127 (6.51)	-0.080 (-6.57)	+0.010 (3.21)
ETH	+0.153 (6.11)	-0.112 (-6.53)	+0.016 (4.68)

Net P&L per unit vega, month fixed effects, standard errors clustered by week. What §7 read as a maturity confound was two weekend effects of opposite sign, which a Friday-noon entry forces to move together and therefore cannot distinguish. Maturity does carry an effect of its own — a day is

worth about one P&L point — but it is an order of magnitude smaller than either weekend coefficient and does not account for them.

- **Selling weekend content pays.** A contract whose remaining life is weekend-heavy is priced richly relative to what it delivers, and shorting it earns +0.127 per unit of weekend share. This is Paper I §5.1's pricing gap showing up as realized P&L, now with maturity held fixed.
- **Living through the weekend costs.** Every unit of weekend inside the holding window costs 0.080. Holding a short across a Saturday loses money even though the Saturday is quiet.

The second is the surprise, and the contract's own quote explains it. Regressing the change in the contract's implied volatility over the hold on the same variables:

	weekend share of window held	weekend share of life sold	maturity
BTC	+0.219 (17.22)	-0.226 (-11.46)	-0.039 (-13.51)
ETH	+0.255 (16.10)	-0.231 (-8.86)	-0.044 (-13.13)

As the weekend is consumed, the life that remains becomes weekday-heavy, so implied volatility per unit of remaining time has to re-rate upward — and a short pays that on the mark. Holds containing no weekend see implied volatility rise by 0.036 on average; holds that are weekend-heavy see it rise by 0.057, half again as much. The quiet Saturday is real and the gamma saving is real — §9 measures it directly — but the re-rating is larger.

Entry day makes the same point without a model, since with the horizon fixed it is entry day that moves the holding window:

entry, 12:00 UTC	Mon	Tue	Wed	Thu	Fri	Sat	Sun
BTC	-0.024	-0.039	-0.037	-0.012	+0.011	-0.053	-0.065
ETH	-0.069	-0.061	-0.047	-0.018	-0.002	-0.087	-0.074
weekend in window	0.00	0.00	0.00	0.00	0.67	1.00	0.33
weekend in life sold	0.00	0.00	0.05	0.30	0.51	0.51	0.17

Saturday entry — a thirty-six hour hold that is entirely weekend — is the worst day in both books and by a wide margin, winning on 28% of trades against 54% for a Friday entry. Friday is the best day in both, and the only positive cell anywhere in the table.

What this implies, and where the raw cells refuse to cooperate. The two coefficients point in opposite directions, so the combination they favour is to sell weekend content without living through it: short a weekend-heavy contract and be out before the Saturday. Thursday entry is the closest a fixed thirty-six-hour horizon allows — weekend share of life 0.30, weekend share of window 0.00 — and it is the best of the four weekday entries, at -0.012 and -0.018, but it is not positive and it is not better than Friday, which does hold the weekend. **The regression separates the two effects; the entry-day table cannot, because entry day moves both at once along with the maturities on offer.** The previous draft read this table as confirming the combination, on a sample a third the size in which the Thursday cell happened to be positive. It should not have been read that way, and the honest statement is that the implication rests on the coefficients alone. Testing the combination properly means letting the holding period vary as well as the entry day; §9 does that, and both coefficients survive with the horizon controlled.

The within-instant fit is reported and is not the evidence. Absorbing a fixed effect for the entry instant and identifying purely from contracts trading side by side gives +0.032 ($t = 1.09$) in Bitcoin and -0.038 ($t = -0.62$) in Ether for weekend share, with maturity likewise indistinguishable from zero. That is a non-rejection from a weak design rather than a contradiction: only 1.7 and 1.8 distinct expiries clear the filters at the average instant, so within an instant the two regressors are still 80% and 82% collinear and the standard errors are two to three times those of the month-fixed-effect fit. Paper I §5.6's lesson applies to this paper's own tables as much as to the market's: **an underpowered non-rejection is not a null.** The separation above rests on the across-instant variation, where maturity is controlled directly and carries a coefficient far too small to explain either weekend term.

9. Varying the holding period, and which Greek the weekend arrives through

§8 held the horizon fixed at thirty-six hours, which meant the entry day alone moved the weekend content of the window and only a Thursday could deliver weekend-rich content without weekend-rich holding. Freeing the horizon — twelve, twenty-four, thirty-six, forty-eight and seventy-two hours, each entered on every day of the week at every available expiry — gives 21,438 Bitcoin and 17,938 Ether trades and separates a third thing: simply holding for longer.

	weekend share of life sold	weekend share of window held	holding hours	maturity, days
BTC	+0.060 (6.54)	-0.035 (-6.47)	+0.0002 (1.76)	+0.0065 (4.74)
ETH	+0.078 (6.24)	-0.053 (-6.44)	+0.0001 (0.94)	+0.0096 (5.87)

Month fixed effects, clustered by week. **Both weekend coefficients survive with their signs and gain significance**, and the structure §8 identified is intact: selling weekend content pays, living through the weekend costs.

Two things differ from §8's fixed-horizon fit, and both are corrections rather than refinements. **Maturity is not zero here.** A day of extra maturity is worth +0.0065 in Bitcoin and +0.0096 in Ether, at five and six standard errors — over the ladder's interquartile range of maturities that is a real effect, and §8's "precisely estimated zero" does not survive the larger sample. What survives is the weaker and more important claim: maturity does not *account for* the weekend coefficients, which keep their size and significance beside it. **And the horizon itself does almost nothing** once maturity is in the regression — the holding-hours coefficient is insignificant in both books, where the previous draft attributed a small positive to theta. That role has passed to maturity, which is where it belonged.

Splitting each horizon by whether its window spans a weekend:

hold	BTC no weekend	BTC spans one	ETH no weekend	ETH spans one
12h	-0.048 (-14.5)	-0.057 (-19.2)	-0.063 (-14.5)	-0.082 (-17.3)
24h	-0.030 (-6.9)	-0.021 (-7.2)	-0.038 (-8.4)	-0.033 (-8.6)
36h	-0.029 (-5.0)	-0.037 (-10.6)	-0.054 (-8.6)	-0.057 (-10.9)
48h	-0.037 (-4.4)	-0.019 (-4.8)	-0.047 (-5.8)	-0.044 (-8.6)
72h	-0.000 (-0.0)	-0.031 (-5.2)	-0.033 (-2.6)	-0.035 (-6.3)

Every cell now loses, which is what the corrected sample looks like once the moving weekends are back in it, so the levels carry no message. The contrast does. **A twelve-hour hold that spans a weekend is entirely weekend, and it is the worst cell in both books by a wide margin — winning on 25% and 24% of trades against 40% and 39% for the twelve-hour hold that spans none.** Beyond twelve hours the raw contrast becomes unreliable and at 24h and 48h it reverses, because the spanning trades also carry far more weekend content in the life they sell — 0.39 against 0.06 at twenty-four hours — and that pays. Only the regression, which prices the two separately, can hold them apart; the raw split cannot, and this is a case where the cell table is the weaker evidence.

Which Greek the weekend arrives through The P&L of a delta-hedged short decomposes, to second order in the entry Greeks and the terminal moves, into a gamma term, a theta term, and the three volatility terms — vega, volga and vanna — plus a residual that carries the path dependence the endpoint approximation cannot see. Running the same regression on each term separately locates the effect. Coefficients on the weekend share of the window held:

term	BTC	ETH
gamma	+0.0324 (3.61)	+0.0144 (1.04)
theta	-0.0034 (-3.83)	-0.0026 (-2.08)
vega	-0.0829 (-17.58)	-0.1057 (-15.77)
volga	-0.0000 (-1.25)	-0.0001 (-1.37)
vanna	-0.0001 (-0.12)	-0.0003 (-0.29)
residual (path)	+0.0201 (1.92)	+0.0427 (2.73)

The entire cost of holding through a weekend is still the vega term, and it is if anything larger than the previous draft reported. Volga and vanna remain economically invisible. The terms sum to -0.034 and -0.052, which reconciles with the -0.035 and -0.053 of the headline regression.

But gamma is no longer nothing, and that is the correction. The previous draft reported a gamma coefficient indistinguishable from zero and put the whole quiet-weekend credit into an unexplained residual. With the moving weekends restored to the sample, the credit appears where theory says it should: the gamma term itself is **positive**, at +0.032 and three and a half standard errors in Bitcoin. Holding a delta-hedged short through a weekend saves gamma cost, because the weekend is quiet — which is the finding this entire paper is about, now visible in the Greek that is supposed to carry it. The residual shrinks accordingly, from +0.036 to +0.020, and in Bitcoin it is no longer significant at conventional levels. A large systematic residual was a sign the decomposition was missing something; it was, and what it was missing was the weekends where the index moved.

Adding the entry Greeks to the headline regression asks whether the weekend measures are proxying for risk exposure. They are not — both coefficients strengthen — but the answer to *which* Greek matters has reversed:

	weekend life	weekend held	gamma/vega	volga/vega	vanna/vega
BTC	+0.072 (7.96)	-0.030 (-5.45)	-0.019 (-12.49)	+0.001 (0.75)	-0.000 (-0.12)
ETH	+0.084 (6.92)	-0.041 (-5.27)	-0.023 (-9.79)	-0.004 (-2.46)	-0.001 (-0.88)

Greek coefficients are shown per standard deviation of the exposure, which is the only way they are readable: gamma per unit vega has a standard deviation of 315 in Bitcoin and 250 in Ether, so its raw coefficient of -0.00006 is a large effect wearing a small number. **Gamma per unit vega is the exposure that matters**, at twelve and ten standard errors and about two P&L points per standard deviation — comparable to the weekend-held coefficient itself. Volga, which the previous draft identified as the only Greek with independent signal, is insignificant in Bitcoin and worth -0.004 per standard deviation in Ether; statistically present there, economically nothing. The reversal has the same cause as everything else in this section: gamma exposure only pays off differently across contracts when the index actually moves, and the earlier sample had removed the occasions when it did.

What this says about entry and exit

- **Entry: choose weekend content, and avoid gamma per unit vega.** The weekend share of the life being sold is worth +0.060 to +0.084 per unit. Among contracts of equal weekend

content, the ones with least gamma per unit vega are preferable — which in practice means longer-dated, consistent with the positive maturity coefficient in the same regression.

- **Exit: before the weekend is consumed, not after.** The cost is not the weekend's realized movement — that is in the gamma term, and it is a *credit*. The cost is the mechanical re-rating of implied volatility as weekend-heavy remaining life turns into weekday-heavy remaining life, and it is paid on the mark whether or not anything happens. §6's corrected exit ladder makes the same point from the other side: holding into Monday, when the re-rating is complete, is the worst of the four exits by a wide margin.
- **Or hedge the re-rating instead of avoiding it.** A long position in a weekday-heavy contract of similar vega gains when implied volatility re-rates upward, which is what the calendar spread of §2 is, and it is a reading of why the spread's Sharpe survived costs that the outright short's did not. This is an interpretation the decomposition suggests, not one it tests: the spread was never run through this attribution.

Two limits. The horizons overlap — a Monday 48-hour trade and a Tuesday 24-hour trade cover some of the same hours — so the effective number of independent observations is well below the nominal count; the week clustering absorbs some of this but not all of it. And the attribution uses entry Greeks with terminal moves, so everything path-dependent lands in the residual by construction. The residual is smaller than it was and no longer significant in Bitcoin, which is a better sign for the decomposition than the previous draft's large one, but it remains a reported column rather than a discarded one.

10. The same rule across the moneyness ladder

Everything from §6 to §9 ran on contracts within a 0.35–0.65 delta band — the most liquid part of the surface and the part where vega is largest. The obvious question is whether the trade is better somewhere else on the ladder: deep in the money, in the money, at the money, out of the money, deep out of the money. The rule is held fixed at the one §6 settled on — enter Friday 12:00 UTC, exit 00:00 UTC on Sunday, re hedge hourly — and only the strike moves.

The design controls maturity in the regression, not in the sampling. An earlier version pinned maturity by hand, requiring all five buckets to print on one common expiry and to print again near the exit. That is the tightest possible control and it starved the wings: 266 Bitcoin and 63 Ether trades in total, with Ether's deep-in-the-money bucket at *one*. The version reported here keeps every contract tradeable at the entry instant and controls maturity with a Friday fixed effect plus each contract's own maturity and weekend share, so a bucket is only ever compared against contracts that were tradeable at the same instant, on the same index, under the same weekend. That gives 18,794 Bitcoin and 14,576 Ether trades across 479 of 496 Fridays.

What actually trades Before asking what each bucket pays, it is worth asking what size each bucket can absorb. Over the whole tape:

	BTC trades	share	share of vega	notional	median premium
deep ITM	235,583	1.9%	0.8%	\$9.4bn	\$3,910
ITM	496,214	4.0%	4.3%	\$23bn	\$1,727
ATM	3,836,460	31.0%	43.6%	\$235bn	\$841
OTM	3,893,315	31.4%	35.3%	\$291bn	\$348
deep OTM	3,929,695	31.7%	16.0%	\$289bn	\$81

Ether's shares are within a percentage point of these throughout. **The two in-the-money buckets together are 6% of trades and 5% of vega.** That is not a quirk of this sample; it is what an options market does. A trader wanting in-the-money exposure buys the out-of-the-money option on

the other side of the same strike, which by put-call parity is the same delta-hedged position and is cheaper to cross. The deep-in-the-money bucket is therefore a bucket in name only, and the analysis below treats it as unmeasurable rather than as a result.

Effective half-spreads, recovered from the aggressor sides, price that illiquidity: 0.33 volatility points at the money in Bitcoin against 0.45 deep out of the money and 0.65 deep in the money, and 0.51 against 0.62 and 1.15 in Ether. **Since the clock trade crosses twice, the wings pay roughly 1.4 times the at-the-money cost before anything else happens.**

What each bucket pays

	BTC n	net/vega	t	hit	cost	ΔIV , vol pts
deep ITM	290	-0.111	-1.78	37%	0.092	+5.9
ITM	1,055	-0.040	-3.96	42%	0.037	+2.1
ATM	5,350	-0.012	-6.29	45%	0.028	+0.0
OTM	5,586	-0.020	-13.39	40%	0.029	+0.9
deep OTM	6,513	-0.045	-16.28	31%	0.041	+3.3

Ether traces the same shape one notch lower throughout: -0.022 at the money, -0.028 out of the money, -0.063 deep out of the money. **Every bucket loses on this contract menu**, which is not a contradiction of §6 — that section selects the single most weekend-heavy contract each Friday, while this one keeps everything out to fourteen days, and the maturity coefficient below is positive. The levels are not the point. The shape is: **at the money is the best place to run this trade and both wings are worse, monotonically in distance from the money.**

The last column says why, and it is the mechanism of §8 sorted by strike. The weekend re-rating of implied volatility is almost invisible at the money, at four hundredths of a volatility point, and reaches **3.3 points deep out of the money**. A short pays that on the mark. The wings are where the smile moves when the weekend is consumed.

Within a Friday, with maturity controlled

relative to ATM	BTC β	t	ETH β	t
deep ITM	-0.046	-0.66	-0.582	-1.88
ITM	-0.007	-0.85	-0.011	-0.94
OTM	-0.009	-4.19	-0.010	-3.36
deep OTM	-0.027	-3.92	-0.032	-3.33
maturity, days	+0.0013	2.58	+0.0024	2.09
weekend share of life	+0.031	1.22	+0.045	0.95

Friday fixed effects, clustered on the Friday. The out-of-the-money penalty survives the tightest control available and replicates across books. As a single continuous statement, distance from the money is worth **-0.086 ($t = -3.66$)** in Bitcoin and **-0.183 ($t = -3.23$)** in Ether per unit of $|\Delta - 0.5|$.

A limit that cannot be resolved with this data. §9 found gamma per unit vega to be the exposure that separates contracts, and volga to be the one whose sign the weekend mechanism predicts. Both are functions of moneyness: the correlation between distance from the money and volga per unit vega is **+0.81** in both books. Adding volga and gamma to the regression duly kills the moneyness coefficient — Bitcoin -0.024 ($t = -0.61$), Ether +0.611 ($t = 1.12$) — but volga and gamma are not significant either once moneyness is beside them. **The three cannot be separated here, and the honest statement is that the wing penalty and the wing convexity are the same fact seen twice, not that one explains the other.**

Put-call parity, which is the test that matters A delta-hedged short call and a delta-hedged short put struck at the same price are the same position: what differs between them is linear in the forward and is exactly what the hedge removes. So the in-the-money and out-of-the-money buckets are two views of the same strikes, and if calls and puts disagree within a bucket, the disagreement is about which contract prints, not about the risk being held.

	BTC calls	BTC puts	ETH calls	ETH puts
deep ITM	-0.204 (-3.02)	+0.031 (0.26)	-0.111 (-0.88)	-1.579 (-3.61)
ITM	-0.050 (-4.30)	-0.026 (-1.39)	-0.068 (-5.49)	-0.066 (-3.80)
ATM	-0.015 (-6.02)	-0.008 (-2.87)	-0.025 (-8.88)	-0.019 (-5.67)
OTM	-0.018 (-9.86)	-0.021 (-9.21)	-0.023 (-9.37)	-0.034 (-11.90)
deep OTM	-0.041 (-12.61)	-0.050 (-11.04)	-0.044 (-17.30)	-0.081 (-12.25)

Parity holds everywhere it can be measured and fails exactly where the volume table says it must. In the at-the-money, out-of-the-money and deep out-of-the-money buckets calls and puts agree to within a percentage point or two of vega — different contracts, same risk, same answer, which is a non-trivial internal check on the whole engine. In the deep-in-the-money bucket they disagree by 0.23 in Bitcoin and 1.47 in Ether, on 115 and 105 put observations against a bucket that is 0.8% of market vega. **That bucket is noise, and its headline -0.111 and -0.740 should not be quoted as findings.**

Verdict on the ladder **There is no better place on the ladder to run this trade.** The at-the-money bucket is where the vega is, where the crossing cost is lowest, where the weekend re-rating is smallest, and where the P&L is least bad; every step away from it costs both in spread and in re-rating, and the two wings are penalised alike once the position is delta-hedged, exactly as parity requires. The in-the-money side cannot be traded at size at all — 5% of vega, the widest spreads on the board, and calls and puts that do not agree. Trade sheets for every trade in both books are saved in `w51_moneyness_sheet_{cur}.csv`.

11. What it is worth

Per-vega figures are the right unit for comparing trades and the wrong unit for deciding whether to run one. This section converts.

Per contract. At the median vega of the contracts the \$6 rule actually selects — \$859 in Bitcoin and \$55 in Ether — the full-history edge of +0.043 and +0.038 per unit vega is:

	median vega	edge/vega	per contract	trades/yr
BTC	\$859	+0.043	+\$37	~27
ETH	\$55	+0.038	+\$2	~22

Capacity. The Friday-noon window — every print within two hours of the entry hour, across the whole tape — carries \$538m of at-the-money vega in Bitcoin over 496 Fridays, or about **\$1.1m of vega per Friday**. A participant taking a realistic 5–10% of that flow without moving the market against themselves is working \$55–110k of vega per weekend, which at +0.043 is \$2,400–4,700 a weekend, or of the order of **\$60,000–120,000 a year gross** across roughly twenty-seven tradeable weekends. That is before staff, data, colocation, the funding paid on the perpetual hedge, and the operational risk of a position that must be delta-hedged hourly through a weekend.

And the recent record is flat. The full-history figure is an average over a sample in which the first half worked and the second did not. Year by year, on the broad near-the-money menu at hourly rehedgeing:

	2021	2022	2023	2024	2025	2026
BTC	-0.023	-0.009	+0.025	+0.004	-0.017	-0.044
ETH	-0.026	-0.036	-0.016	-0.009	-0.001	-0.041

Pooled over 2025–26 that is **-0.028 (t = -10.1)** in Bitcoin, winning on 29% of weekends in 2026. The narrower rule of §6 — one most-weekend-heavy contract per Friday rather than the whole near-the-money menu — holds up considerably better and still converges on nothing: **-0.001 (t = -0.12)** in Bitcoin and **+0.007 (t = 0.32)** in Ether over the same two years.

A trade whose expected value is indistinguishable from zero, whose capacity is six figures, and whose worst historical weekend costs 0.72 per unit vega, is not a business. It is a measurement. Note that the figures in this section are for the outright clock trade of §6, which is a single short leg and so is unaffected by the costing correction; the calendar spread, charged correctly, does not reach this section's arithmetic at all because it never clears zero. §12 prices the one configuration that does.

12. The only configuration that clears zero

Sections 3 to 11 establish that the signal is real and that no *taker* construction covers its costs. Since the binding constraint turns out to be the toll rather than the edge, the question worth asking last is what it would take to stop paying the toll. There is one answer available on this venue: quote rather than cross.

A market maker running the same position differs from a taker in exactly one term. The effective half-spread — recovered from the tape by differencing buyer-paid against seller-received implied volatility on the same instrument-day, 0.42 volatility points in Bitcoin and 0.63 in Ether — is paid by the taker on both legs and *earned* by the maker on both. The swing is four half-spreads. Everything else is identical: the same contracts, the same delta hedge, the same exchange fees, the same weekend.

daily reheding, per unit vega	BTC	ETH
gross spread	+0.0418	+0.0418
exchange option fees, both legs	-0.0297	-0.0295
perpetual hedging fees, both legs	-0.0282	-0.0286
spread crossing, both legs	-0.0083	-0.0127
taker net	-0.0245	-0.0289
maker net (earns the spread rather than paying it)	-0.0079	-0.0036

Making markets closes two thirds of the gap and does not cross zero. That is the single most useful number in this paper for anyone deciding whether to build something here, and it is worth being precise about why. The bid-ask was never what killed this trade. Of the 0.066 a Bitcoin taker pays, the spread is 0.008; the other 0.058 is fees, and fees fall on the maker exactly as they fall on the taker.

So the decision reduces to the fee schedule. Splitting that 0.058 — by zeroing each fee function in turn and re-walking the same hedge path, rather than re-deriving the arithmetic — gives 0.030 of exchange option fees and 0.028 of perpetual hedging. Only the first is negotiable. A maker paying no option fee at all would earn:

	BTC	ETH
maker, standard fees	-0.0079	-0.0036
maker, zero option fee	+0.0218	+0.0259
option-fee discount needed to break even	27%	12%

Those are the first positive net figures in this paper. A desk quoting both legs passively, hedging daily, and paying 27% less than the standard option fee in Bitcoin — 12% in Ether — would have made money on this trade across the full history, at roughly +0.022 per unit vega. Deribit's published schedule does tier by volume, so this is not a hypothetical threshold; it is a threshold a large enough participant could actually be on.

And it does not survive the recent period. Over the last twelve months the maker version earns -0.043 in both mature books, and adding back the entire option fee still leaves about -0.013. The configuration that clears zero clears it on history, in a window during which §3.1 shows the gross signal was already narrowing. There is no twelve-month window at the end of this sample in which any construction tested here — taker or maker, spread or outright, at any rehedging frequency, on any part of the smile — makes money.

Three caveats bound how far this can be pushed, and they run in both directions.

- **Passive fills are assumed, not modelled.** A maker quoting both legs does not get filled on demand; they get filled when someone crosses, which is disproportionately when the crosser is right. That adverse selection is a real cost this calculation omits entirely, and it can only make the maker number worse. The trade tape carries the aggressor side, so it is measurable; it is not measured here.
- **The half-spread is an average.** It is estimated per instrument-day from both-sided flow, so it describes the contracts that trade both ways, which are the liquid ones. A maker's realised capture on a thin weekend-heavy contract could be wider or could be zero.
- **A maker has inventory a taker does not.** The calculation prices one round-trip in isolation. A real book warehouses vega between fills, and Paper I §5.6 shows the weekend's variance distribution has been getting more tail-heavy since 2022 — which is precisely the risk a warehousing maker carries and this arithmetic ignores.

The honest summary is that this paper has found the one configuration in which the trade is not obviously dead, and that configuration is dead over the last year, rests on an assumption about fills it does not test, and would need a fee tier to have worked even on history. **That is not a business. It is a well-measured reason to stop looking.**

13. Conclusion

This paper set out to harvest the pricing error documented in Paper I and did not succeed. What it found instead is worth separating into three claims of descending confidence.

The error has never been harvestable by crossing, and the gross signal is fading besides. The calendar spread's gross edge of +0.042 per unit vega is real and tracks the pricing gap across four books in the right order, but the venue charges 0.066 to collect it; an earlier version of this paper reported a Sharpe of 1.5 and an inversion, both artefacts of a spread that credited its long leg's costs instead of charging them. The outright clock short, which admits no such error, earns +0.043 over the full history and nothing over the last two years. So there are two facts, not one: costs have always exceeded the edge, and the edge has been shrinking anyway.

The mechanism is now well identified, and it is a re-rating rather than a realized-variance effect. Selling weekend content pays; holding through the weekend costs; the two have opposite

signs and a Friday-noon entry cannot tell them apart. What makes the second true is that implied volatility per unit of remaining time must rise as the weekend is consumed, and the short pays that on the mark whether or not anything happens. The second-order attribution puts the entire weekend cost in the vega term, and finds the quiet weekend itself arriving as a *credit* in the gamma term — the paper’s central fact showing up in the Greek that should carry it. §9 is the strongest part of this paper and it is the part with no trading implication at all.

The residual pricing error is probably being quoted against rather than traded, and §12 now puts a number on how far that gets anyone. A maker earns the half-spread on both legs instead of paying it, which is worth 0.017 per unit vega in Bitcoin and 0.025 in Ether and still leaves the position short of zero at standard fees. Only a fee tier discounting option fees by 27% and 12% respectively turns it positive, at +0.022, and only on full-history data — the last twelve months lose about 0.013 even then. A desk that prices the weekend clock correctly still improves its marks on inventory it holds anyway, at no crossing cost and no capacity limit, and that remains the most likely reason the quote-side gap Paper I measures survives. But the maker arithmetic here also omits adverse selection entirely, which can only make it worse.

What this leaves for someone else. The trade is dead at taker capacity and marginal at maker capacity, but the measurement is not, and the asymmetry is the point: an error that survives in quotes while being unharvestable by crossing is the equilibrium a market maker would produce, not the one an inefficient market would. The next measurement is the one §12 names and does not make — adverse selection on passive fills, which the aggressor side of this tape can support and which would settle whether the maker configuration is genuinely marginal or merely appears so because the cost of being filled has been left out.

A closing note on the correction. This draft’s headline result reversed because of a two-line construction error that no test caught, since each leg was individually right and only their combination was wrong. The check that would have found it is the one now in `tests/test_spread_costing.py`: costs add across the legs of a spread, they do not cancel. It is worth stating as a rule rather than a fix — **whenever a strategy’s P&L is assembled by differencing two separately-costed positions, the differencing silently reverses the sign of one side’s costs.** The engine was correct throughout; the assembly was not.

Reproduction

Every table in this paper is regenerated by the scripts below, in order. The measurement chain they depend on is Paper I’s; see its reproduction table.

What	§	Script	Tables
Spread P&L, date by date	3, 4	<code>weekend_short.py</code>	<code>w32-w35_*.csv</code>
Outright short, rehedged ladder	5	<code>weekend_short.py</code>	<code>w36-w37_*.csv</code>
Corrected costing, maker economics	4, 12	<code>weekend_maker.py</code>	<code>w60-w61_*.csv</code>
Fixed-hour entry and exit	6	<code>weekend_clock.py</code> <code>--grid</code>	<code>w38-w43_*.csv</code>
Rehedge ladder by exit	6	<code>weekend_rehedge_exits.py</code>	<code>w41b_*.csv</code>
Entry filters, pre-specified	7	<code>weekend_filters.py</code> <code>--wide</code>	<code>w44-w45_*.csv</code>
Weekend content vs maturity	8	<code>weekend_content.py</code>	<code>w46-w48_*.csv</code>
Hold ladder, Greek attribution	9	<code>weekend_content.py</code> <code>--hold-ladder</code>	<code>w49-w50_*.csv</code>
Moneyness ladder, volume, sheets	10	<code>weekend_moneyness.py</code>	<code>w51-w54_*.csv</code>

Trade sheets for every trade in §10 are saved per currency in `w51_moneyness_sheet_{cur}.csv`, one row per trade with both timestamps, both marks, the hedge P&L, every fee component and the second-order attribution.

Open items

- ☐ **When exactly did the trade die, and against what?** The year-by-year path in §11 shows the turn but not its cause. The candidates — the arrival of systematic market makers, the 2024 spot ETFs, the growth of weekend perpetual liquidity — are all datable, and the P&L series is long enough to support a break test that this draft does not run.
- ☐ **Is the quote-side error being maintained by the same desks that killed the trade?** §12's central inference is untested. It needs quote-level data or maker-taker identifiers, neither of which the public trade tape carries.
- ☐ **Separate the wing penalty from wing convexity.** §10 finds distance from the money costs -0.086 and -0.183 , and cannot tell that apart from volga per unit vega, with which it correlates at $+0.81$. A cross-section with independent variation in the two — a wider maturity span, or a venue with a different smile — would settle it.
- ☐ **The overlapping-horizon problem in §9.** A Monday 48-hour trade and a Tuesday 24-hour trade share hours, so the effective number of independent observations is well below the nominal count. Week clustering absorbs some of this and not all of it; a block bootstrap over non-overlapping windows would bound it.
- ☐ A reference list. The draft cites informally throughout and has no bibliography.